https://doi.org/10.1038/s44323-026-00088-y

Evaluation of wearable light dosimeters for circadian lighting: spectral, spatial, photometric, melanopic, and thermal performance

Check for updates

R. S. Kore[1,2] ✉, V. Boyer[1], D. A. Reichenberger[1], J. Mundinger[1], A. M. Chang[1] & D. Durmus[1]

Light is the primary stimuli for the entrainment of human circadian rhythms and holds enormous potential in aiding precision medicine. Laboratory studies highlight the roles of photic history, intensity, spectrum, and timing, but field translation gets complicated by real-world confounds. Measuring daily light exposure in realistic conditions requires an accurate, durable, and affordable wearable light dosimeter. Here, the spectral, spatial, and thermal performance of three wearable dosimeters (LYS Buttons, Blue Iris Specks, and Actiwatch Spectrum Plus) are investigated and benchmarked against a calibrated spectroradiometer under multipole lighting conditions. Light measurements taken by dosimeters under various lighting conditions are then used to calculate melanopic and photometric measures. The LYS Button was the easiest to use, albeit with reduced accuracy. The Blue Iris Speck had the highest photometric and data-logging accuracy. The Actiwatch had the poorest photometric performance, which raises caution for scientific use. The results inform device development and protocol design for chronobiology and sleep research. Future studies should evaluate and compare the performance of the dosimeters based on body-worn position.

Light is fundamental not only for vision but also for non-image forming (NIF), or alternatively integrative[1], effects on occupants that regulate physiological and psychological health. Beyond enabling sight, light exposure influences physiology (e.g., sleep–wake timing, hormone release, and metabolism)[2–7], and behavioral outcomes, such as alertness, mood, and cognitive performance[8–13]. Disruptions in light exposure have been associated with medical and mental health conditions, such as seasonal affective disorder (SAD)[14], fragmented sleep patterns in individuals with Alzheimer's disease[15,16], and developmental challenges in premature infants[17,18]. Understanding and quantifying these effects is crucial for designing architectural environments that support occupant health.

The regulation of circadian rhythms is primarily governed by the melanopsin photopigment[19], located in intrinsically photosensitive retinal ganglion cells (ipRGCs)[20]. This specialized phototransduction pathway enables the human body to synchronize with the external light-dark cycle, thereby influencing sleep-wake patterns, hormone secretion, and overall well-being[14,20]. Extensive research has characterized the spectral sensitivity of melanopsin, providing a deeper understanding of its physiological role. Studies have determined its peak sensitivity and response to different wavelengths[21], leading to the development of light exposure metrics designed to assess circadian-effective light.

To translate these health-related impacts and scientific insights into practical applications, several metrics have been proposed[22–25] that often provide conflicting results[26,27]. More recently, a set of circadian metrics was standardized based on consensus to quantify light exposure's biological effects[22]. These circadian lighting metrics serve as essential tools in evaluating light exposure, its potential effects on occupants, and guiding the design of illuminated architectural environments that promote human health and performance.

Controlled laboratory studies often aim to validate circadian metrics by demonstrating correlations between light exposures and physiological outcomes. While controlled lab studies can provide strong evidence for the relationship between light exposure, circadian regulation, and specific health outcomes, utilizing these metrics in field studies faces additional complexities. Unlike controlled laboratory environments, real-world light exposure is highly variable, influenced by personal exposure patterns, environmental lighting conditions, and individual physiological differences[28,29]. Accurately measuring light exposure in these contexts requires high-quality, wearable dosimeters worn as close to the eyes as

[1]Pennsylvania State University, University Park, PA, USA. [2]Virginia Tech University, Blacksburg, VA, USA. ✉e-mail: rugved@vt.edu



Fig. 1 | Light-measurement devices used in the study. This figure shows the dosimeter Blue Iris Speck and LYS Button, reference spectroradiometer Gigahertz S-BTS256, and dosimeter Actiwatch Spectrum Plus.

possible (e.g., light dosimeter (Lido[30])), that are capable of accurately capturing light exposure and calculating circadian-relevant metrics[30–32].

Several light dosimeters of varying capabilities in measuring circadian-effective light have been developed to address the urgent need to collect more accurate data. Differences in spectral sensitivity, response accuracy, and performance in dynamic lighting conditions necessitate a systematic evaluation of these devices[33–35]. This study aims to compare the performance of three prominent wearable light dosimeters under different lighting conditions, including daylight. The evaluation focuses on key performance characteristics, such as output linearity across illuminance levels, angular response, and temperature response. The findings aim to contribute to the refinement of circadian light measurement techniques and aid in selecting appropriate devices for use in sleep and circadian photobiology research.

The circadian rhythm, our internal timekeeping system, is closely tied to the daily light-dark cycle. Within this intricate system, melanopsin, a photosensitive pigment expressed in ipRGCs, acts as the primary photoreceptor, transmitting signals to the master circadian clock suprachiasmatic nucleus (SCN)[19,36]. The melanopsin photopigment itself has a peak spectral sensitivity of ~479 nm, which shifts to ~490 nm in the human eye after preceptoral filtering by the lens and macular pigment[21,37–39]. The interplay of distinct cone photoreceptors, classified as short (S), medium (M), and long (L) wavelength-sensitive cones, also contributes to circadian rhythm regulation[40], though their contributions are context-dependent and probably secondary to ipRGC-mediated pathways. Many studies indicate that ipRGCs are the primary driver of non-visual responses under most lighting conditions[41–47]. Recent studies have underscored the influential role of these cone types in mediating non-visual responses to light, shedding light on their contribution to the intricacies of the human biological clock[1,48–50]. The photopic luminous efficiency function, developed by cone responses and responsible for vision at photopic levels, has peak sensitivity at longer wavelengths (555 nm)[51,52]. Therefore, devices measuring light to study the human vision and body clock need to account for the differences in spectral responses of cone, rod, and melanopsin photoreceptors. It is also crucial to thoroughly document the design and calibration of dosimeters to enable users to contextualize the results and their accuracy.

The spectral power distribution (SPD), spatial distribution, photic history, intensity, duration, and timing of light exposure, along with their cumulative effects, significantly impact melatonin production and, consequently, the human circadian rhythm[53–55]. In daily life, individuals encounter a diverse array of lighting conditions at various times, necessitating in-situ measurement of these parameters to comprehensively understand their overall impact on the human circadian system. While numerous laboratory studies have investigated this relationship, they often focus on isolated parameters or controlled timeframes, typically measuring light exposure on a vertical plane using portable dosimeters[56]. This approach inadequately represents real-life scenarios where individuals experience dynamic light exposure patterns. Employing a high-accuracy wearable dosimeter, particularly at eye-level, with ample battery life and data storage capacity, can provide a more accurate and comprehensive representation of an individual's light exposure in naturalistic settings[57,58].

In recent years, many research labs and private companies have developed commercial and research-grade circadian light dosimeters- a recent study identified more than 50 dosimeters[35], some capable of measuring blue light[59], but their accuracy is not always independently tested. The position in which the device is worn on the body, the type of light-sensitive cells, data-logging, and cost are the main variables separating these wearable light dosimeters[31]. LYS Button, Blue Iris Speck, and Actiwatch Spectrum are some of the commonly used light dosimeters in research studies.

The LYS Button (LYS Technologies, Denmark) is a compact wearable dosimeter that provides output based on three-channel sensors (RGB). It is a button-shaped device with a 24 mm diameter and 12 mm height that can be clipped on the outermost layer of the participant's clothing at chest-level or higher to measure the light closest to the eyes. It has a V-shaped transparent optical window, and the entire unit sits in a rechargeable portable battery pack when not in use, as shown in Fig. 1. Due to its light weight, the device causes little to no disturbance to the participant[60] yet can be easily lost or discarded. LYS has been, and still is, widely used in several research studies due to its lower price, compact size, and connectivity to the LYS smartphone application[61,62].

The second dosimeter, Blue Iris Speck (Blue Iris Labs, USA), is a small and low-cost seven-channel spectral dosimeter that outputs reconstructed SPD of the light source between 350 nm and 770 nm (reported in 5 nm intervals). The Speck can be worn as a clipped dosimeter to clothing or as a pendant, or the device can be mounted on walls. These Specks have a smooth, curved, diffused optical window that blends with the housing (Fig. 1). This device is gaining more attention within the research community[35,59,63].

Finally, Actiwatch (Cambridge Neurotechnology, Fenstanton, Cambridgeshire, UK) is a wrist-worn activity logger developed by Cambridge Neurotechnology, with the subsequent sale of the trade rights to Philips Respironics. While not all Actiwatch models include light sensors, certain versions, such as the Actiwatch Spectrum and Actiwatch Spectrum Plus, are equipped with integrated light sensors and have been widely used in research to measure personal light exposure. As the name suggests, it is designed in the watch-like form factor with a three-channel RGB photosensor located in a recessed optical well (Fig. 1). Few studies have already characterized the Actiwatch Spectrum[64–69] and Actiwatch Spectrum Plus[70].

Table 1 | Statistical analysis of difference in illuminance measurements from dosimeters in comparison with reference

			Light Dosimeter								
Light source	Statistical Test		Button1	Button2	Button3	Speck1	Speck2	Speck3	Actiwatch1	Actiwatch2	Actiwatch3
LED	Wilcoxon	*W*	631.5	370.5	636	613	613	610	478	620	414
		z-score	−0.116	−4.549	−0.029	−0.475	−0.475	−0.534	−3.095	−0.34	−4.22
		p	0.454	***	0.488	0.317	0.317	0.297	***	0.367	***
	Effect size	*r*	−0.016	−0.686	−0.004	−0.067	−0.067	−0.075	−0.438	−0.048	−0.603
Daylight	Wilcoxon	*W*	176	171	192	235	19	315	171	171	171
		z-score	−4.967	−5.125	−4.461	−3.101	−16.004	−0.569	−5.125	−5.125	−5.125
		p	***	***	***	***	***	0.285	***	***	***
	Effect size	*r*	−0.828	−0.854	−0.744	−0.517	−3.412	−0.095	−0.854	−0.854	−0.854

Studies suggest that Actiwatch devices are reliable for monitoring mobility and collecting derived sleep data, but need to be used with caution when drawing causal relationships between light and other biological processes due to high intra-watch data variance[64,69,71–77]. Thus, a group of researchers developed a test method for field calibration of Actiwatch Spectrum dosimeters, and they suggest that knowledge of inter-equipment variability and calibration of a batch of measuring devices is essential[65].

Spectral accuracy, data storage, battery life, and mount type are the four main aspects of a wearable dosimeter[67,78]. High spectral accuracy, ideally capturing the full spectrum of light, is crucial for calculating reliable alpha-opic metrics, such as melanopic irradiance[59]. Storing these raw and calculated data with accurate timestamps is the next critical step, typically achieved through onboard storage or cloud uploading[79,80]. To ensure smooth data collection, a wearable dosimeter should have sufficient storage and battery life to last at least overnight and preferably 24 h or more[32,81]. For most circadian field studies, multi-day continuous recording (e.g., 7–14 consecutive days) is commonly recommended to capture prior light history and weekday-weekend differences[82–85]. Accordingly, devices with limited onboard storage can still be suitable if they provide robust buffering and reliable upload in real-world use; however, devices that require frequent syncing should be evaluated carefully to avoid systematic data loss during typical multi-day deployments. Finally, to most accurately represent the light entering the eyes, the device should ideally be mounted as close to the eyes as possible on a vertical plane[31], as the wearing position can impact light exposure accuracy and user comfort[86].

Light is a critical independent variable in sleep and chronobiology studies, necessitating careful control and documentation to ensure accurate assessments of its effects[87–89]. Since the spectral response of melanopsin has been well-characterized, various metrics have been developed to quantify its impact on circadian entrainment. Examples include melanopic equivalent daylight illuminance (mel-EDI), and other α-opic EDIs[22], circadian stimulus/circadian light (CS/CLA), and equivalent melanopic lux (EML) used in some building standards. Melanopic illuminance (E_z, unit: melanopic lux), which aimed to quantify the photoreception of melanopsin[37], was one of the earliest metrics introduced for this purpose. Another circadian metric is the circadian stimulus (CS), which quantifies the effectiveness of light in regulating circadian rhythms. CS has a relative scale ranging from 0 (threshold for circadian system activation) to 0.7 (response saturation). It is directly proportional to nocturnal melatonin suppression after one hour of exposure, where a CS value of 0.7 corresponds to approximately 70% suppression of melatonin[23]. Similar to other circadian metrics, CS may provide a standardized approach to quantify light's biological impact, aiding researchers in evaluating the effectiveness of different light sources in supporting circadian health, but it contains inconsistencies due to its built-in blue-yellow opponency function[90] and describes only nocturnal melatonin suppression[91].

Instead, the International Commission on Illumination (CIE) recommends using melanopic irradiance (E_{mel}), since other metrics such as CS, EML, or melanopic illuminance are not SI-compliant[22,92]. The CIE's melanopic irradiance (E_{mel}) is mathematically expressed as:

$$E_{mel} = \int_{\lambda} E_{e,\lambda}(\lambda) s_{mel}(\lambda) d\lambda \tag{1}$$

where:

- $E_{e,\lambda}(\lambda)$ represents the spectral irradiance (W/m²/nm),
- $S_{mel}(\lambda)$ is the spectral sensitivity function of melanopsin,
- λ is the wavelength (nm).

To standardize comparisons across different lighting conditions and provide alternatives relevant to visual functions, melanopic equivalent daylight illuminance ($E_{v,\alpha}^{D65}$; mel-EDI) has also been introduced. Mel-EDI represents the illuminance produced by a light source that conforms to standard daylight (D65) while providing the same melanopic irradiance (E_{mel}) as the test source. This metric further facilitates comparisons between electric and natural lighting conditions in circadian research. It is expressed as:

$$E_{v,\alpha}^{D65} = \frac{E_{\alpha}}{K_{mel,v}^{D65}} \tag{2}$$

where:

- E_{α} is the melanopic irradiance (unit: W/m²),
- $K_{mel,v}^{D65}$ is the melanopic efficacy constant for daylight, with a value of 1.3262 mW/lm.

Results

The results are organized to quantify dosimeter performance relative to the reference spectroradiometer under controlled LED and scene-based daylight conditions. We first report illuminance agreement and within-device variability, followed by spectral-output comparisons and mel-EDI performance (where available). Finally, characterization results were presented for directional response (f_2) and temperature dependence ($f_{6,T}(T)$) to highlight deployment-relevant sensitivities.

Illuminance

To analyze the illuminance and mel-EDI measurement performance, all test dosimeters (three Buttons, three Specks, and three Actiwatch dosimeters) were compared against the reference spectroradiometer (Gigahertz S-BTS25) and then compared against each other within device type to assess variation. Here, accuracy is used as agreement with a reference (systematic bias), while Table 1 reports between-unit variability within device type. A one-sample Kolmogorov-Smirnov test was performed to check the normality of the data. For statistical comparison of illuminance under LED and

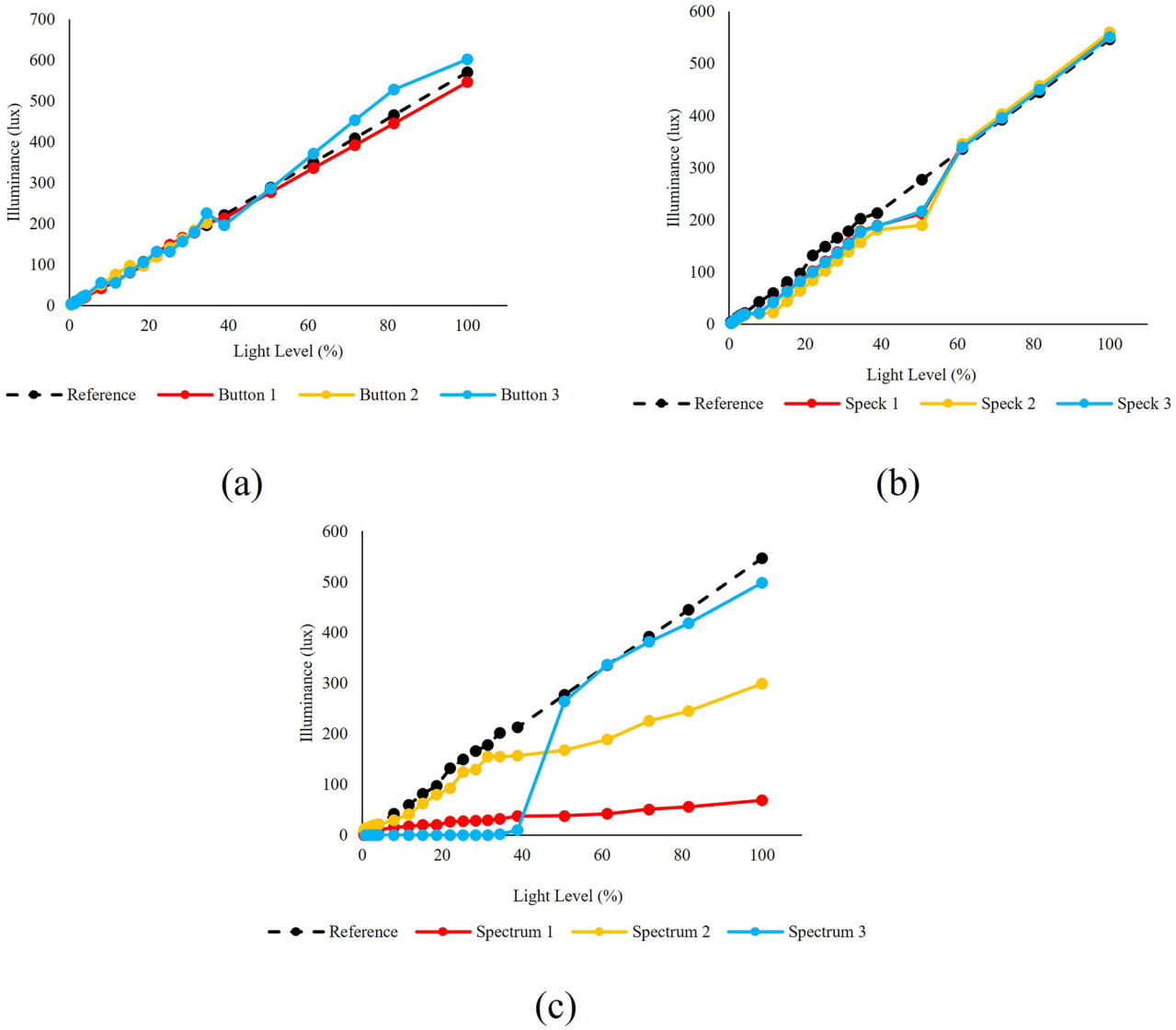


Fig. 2 | Illuminance agreement under LED lighting. Comparison of illuminance measurements under LED lighting at different light levels between reference (black line) and three test light dosimeters (red, yellow, and blue lines) within each type: **a** LYS Button, **b** Blue Iris Speck, and **c** Actiwatch Spectrum Plus.

daylight, spectral measurements under the reference spectroradiometer, originally measured at 1 nm intervals, were first resampled to 5 nm and then calculated the reference illuminance and mel-EDI values.

Illuminance comparison under LED

For all dosimeters, the illuminance and mel-EDI data were not normally distributed. Since non-parametric tests do not assume that the data are approximately normally distributed and they are based on fewer assumptions, the Wilcoxon rank-sum test (significance level $\alpha = 0.05$) was used to assess statistically significant differences between test and reference measurements[93]. No statistically significant differences were observed between the reference and any of the Speck dosimeters. For Button dosimeters, two units did not differ significantly from the reference, while one did. For the Actiwatch dosimeters, only one unit did not differ significantly from the reference, while two did, as shown in Table 1 and Fig. 2. Out of 34 light level measurements, one of the Button dosimeters (Button-2) failed to report six measurements, and one Actiwatch dosimeter failed to report 18 measurements, all due to unknown reasons. A one-way Kruskal-Wallis test (significance level $\alpha = 0.05$) was performed to assess the variance of measurement within each dosimeter type. There was no difference between Blue Iris Specks and LYS Buttons illuminance measurements, although the Actiwatch dosimeters showed a difference between the three Actiwatch devices, as shown in Table 2. These results aligned well with a previous pilot study performed by the authors[94].

Table 2 | Statistical analysis of difference in illuminance measurements within the same light dosimeters

		Dosimeter		
Light source	Statistical test	LYS	Blue Iris	Actiwatch
LED	Kruskal–Wallis test *H*(2)	2.33	0.073	20.601
	Statistical significance (*p*)	0.312	0.964	<0.001
Daylight	Kruskal–Wallis test *H*(2)	6.817	6.620	12.302
	Statistical significance (*p*)	0.033	0.037	0.002

Illuminance comparison under daylight

Performance of all light dosimeters was contrastingly different under daylight. A difference was observed between the reference and all test light dosimeters, except for Blue Iris Speck3 (Table 1; Fig. 3). Buttons and Specks measurements were not statistically different within the respective device

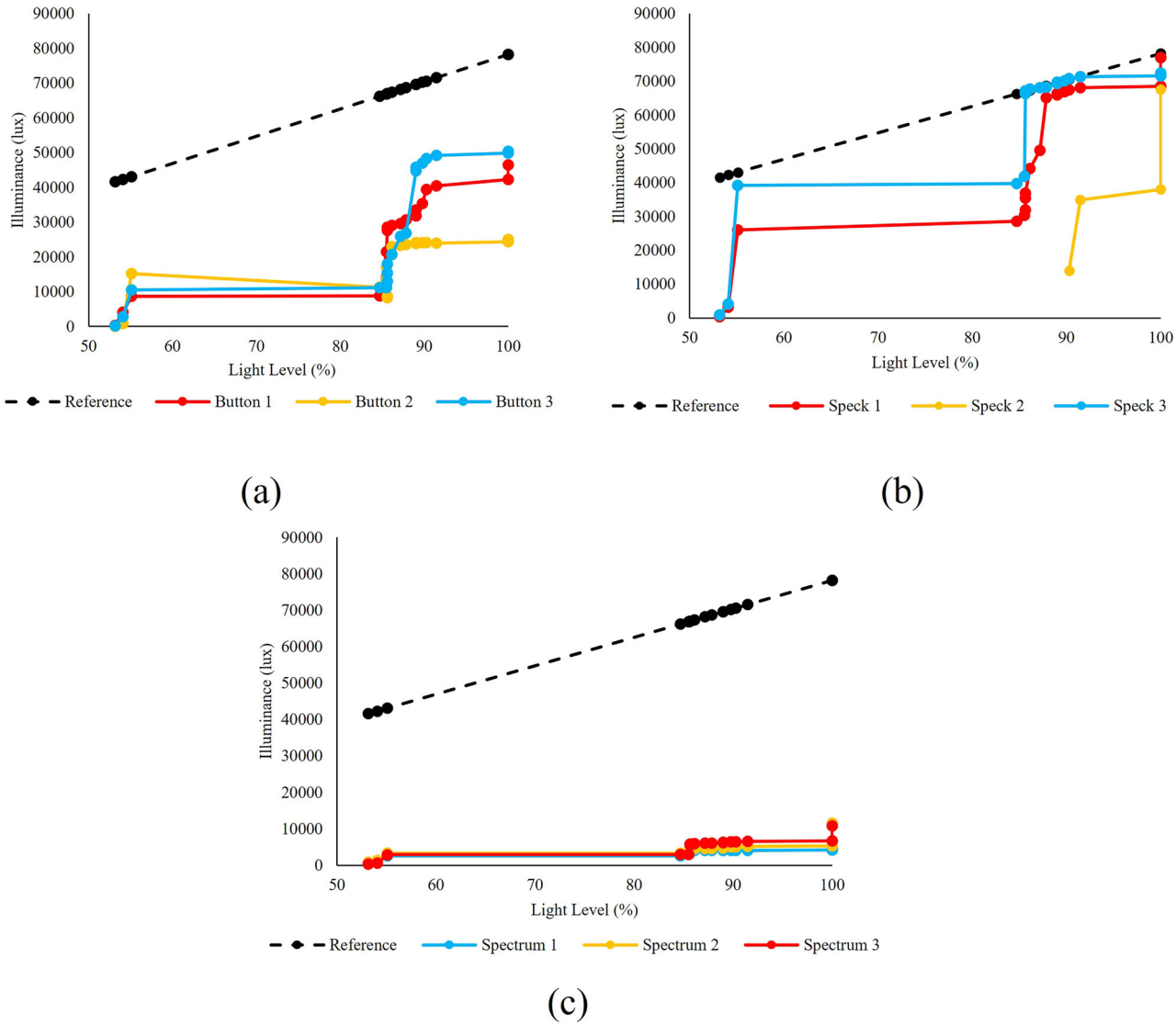


Fig. 3 | Illuminance agreement under daylight. Comparison of illuminance measurements under daylight at different light levels between reference (black line) and three test light dosimeters (red, yellow, and blue lines) within each type: **a** LYS Button, **b** Blue Iris Speck, and **c** Actiwatch Spectrum Plus.

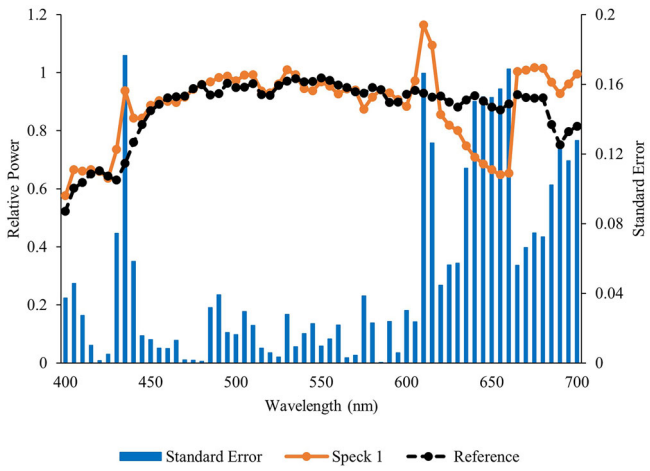


Fig. 4 | Spectral power distribution measurement comparison under daylight. The spectral power distribution of daylight measured using Blue Iris Speck (orange line) and reference spectroradiometer (black line), and the standard error between devices (blue stacked columns).

types, but light measurements from three Actiwatch dosimeters did show differences within the dosimeter type, as shown in Table 2. During this measurement, 14 out of 20 Speck2 measurements failed due to a poor connection with the gateway.

A key limitation of the daylight comparison is that measurements were collected under partly cloudy conditions, where rapid temporal changes in daylight can interact with device-specific integration times and unpublished intra-epoch averaging algorithms[95]. As a result, even time-synchronized samples may reflect non-identical effective exposure intervals across devices, which can bias intercomparisons[96]. Therefore, the daylight results should be interpreted as indicative of field behavior under variable sky conditions rather than as a strict linearity assessment under a stable source.

Spectral output

The spectral measurements from the Speck dosimeter were compared against the calibrated reference spectroradiometer under two lighting conditions: partly cloudy daylight and LED (4600 K). In this section, the spectral output of only one Blue Iris Speck unit was analyzed and compared with reference; thus, this section does not include within-device variability analysis. Under daylight conditions (Fig. 4), the Speck SPD output followed the general shape of the reference curve but exhibited deviations in both

amplitude and spectral detail. The Speck overestimated relative power in the shorter wavelength region (below 450 nm), while underestimations appeared in the middle and longer wavelengths, particularly around 600–650 nm. The standard error was highest between 600 and 700 nm, reaching values above 0.16 in relative units, indicating greater uncertainty in this range.

Under the LED source, spectral differences between the Speck and the reference were more pronounced, as shown in Fig. 5. The spectra measured by the reference device had a smooth curve typical of phosphor-converted (PC) white LEDs, while the Speck output displayed sharper, exaggerated peaks, particularly near 450, 540, and 610 nm. These features suggest the Speck device may be capturing narrowband components less accurately or applying different internal weighting. The standard error across wavelengths remained low in absolute terms (<0.003), but the mismatch in peak shapes and locations indicates limited spectral fidelity under LED lighting. Together, these results suggest that, while the Speck captures the general shape of SPDs under daylight, its accuracy degrades under spectrally complex sources, such as LEDs. Furthermore, the magnitude and distribution of standard errors suggest higher measurement uncertainty at longer wavelengths, particularly under daylight.

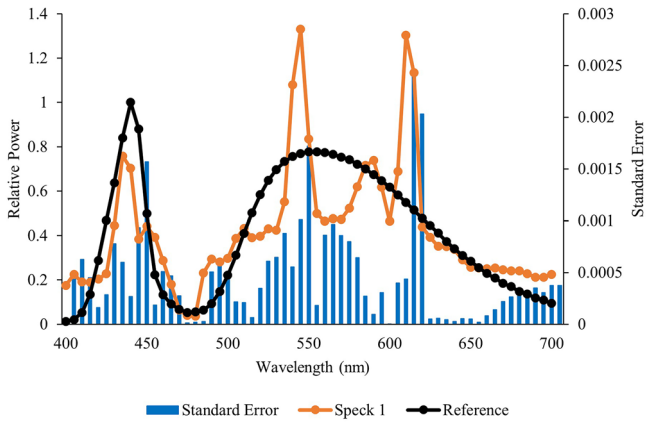


Fig. 5 | The spectral power distribution of 4600 K LED measured using Blue Iris Speck (orange line) and reference spectroradiometer (black line), and the standard error between devices (blue stacked columns).

Melanopic equivalent daylight illuminance (mel-EDI)

To evaluate the performance of wearable dosimeters in measuring mel-EDI, recorded outputs from the Speck and Button dosimeters were compared against a reference mel-EDI standard across a range of reference illuminance levels. The Actiwatch dosimeter was excluded from this analysis, as it does not report melanopic-weighted metrics. This mel-EDI comparison was conducted using the controlled LED protocol—daylight mel-EDI was not evaluated in this study.

The analysis revealed that both Speck and Button dosimeters exhibited strong overall correlation with the reference mel-EDI, particularly at moderate-to-high illuminance levels (200 lx and above). The Speck showed tight clustering and close alignment with the reference across all three units (Fig. 6). At a reference mel-EDI of 498 lx, Speck units reported values ranging from 497 to 537 lx, indicating consistent agreement with the reference at high light levels with minimal within-device variation. The Button dosimeters also tracked the reference reasonably well in high illuminance conditions, albeit with slightly greater variation across units. For example, Button 1 slightly overestimated across the upper spectral range, while Button 3 closely matched the reference. Recorded mel-EDI values ranged from 510 lx to 523 lx for the same reference point of 498 lx.

Distinction between dosimeters became more pronounced at lower illuminance levels (<200 lx). The Speck dosimeters maintained their linearity and relative accuracy, with only minor underestimations observed. For instance, at a reference mel-EDI of 5.9 lx, Speck dosimeters recorded values between 5.7 and 5.8 lx. In contrast, the Button dosimeters demonstrated a slight tendency to overestimate mel-EDI in this region, with values ranging from 6 lx to 7 lx, suggesting reduced sensitivity or calibration discrepancies under dim lighting. Given the emphasis in circadian research on low-light thresholds (e.g., around 1 lx–10 lx mel-EDI), although the protocol included data points in this range, performance near ~1 lx was not explicitly characterized here and warrants dedicated testing.

Overall, both dosimeter types are suitable for high-light level applications. However, the Speck dosimeters outperformed the Button dosimeters in terms of agreement with reference at low-light levels and consistency across units. These characteristics make the Speck dosimeters potentially more reliable for use in circadian lighting studies, particularly those involving dim environments, nighttime exposures, or clinical contexts.

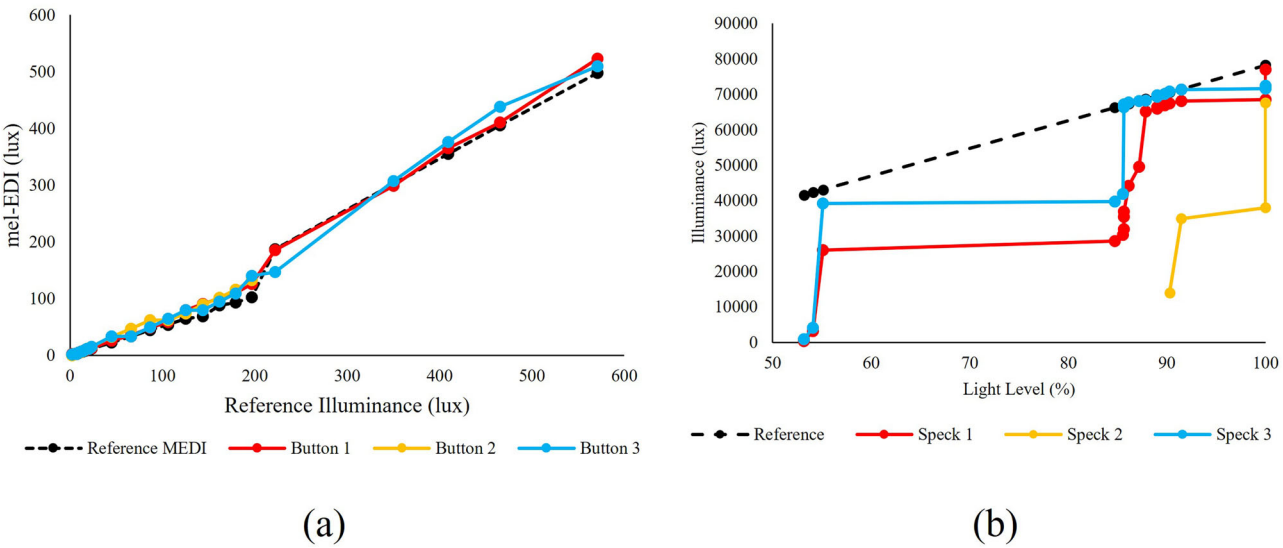


Fig. 6 | Melanopic equivalent daylight illuminance agreement under LED lighting. The mel-EDI comparison under LED at different light levels between reference m-EDI measured using a calibrated spectroradiometer (black line) and three test light dosimeters (red, yellow, and blue lines) within each type: (**a**) LYS Button and (**b**) Blue Iris Speck (left figure).

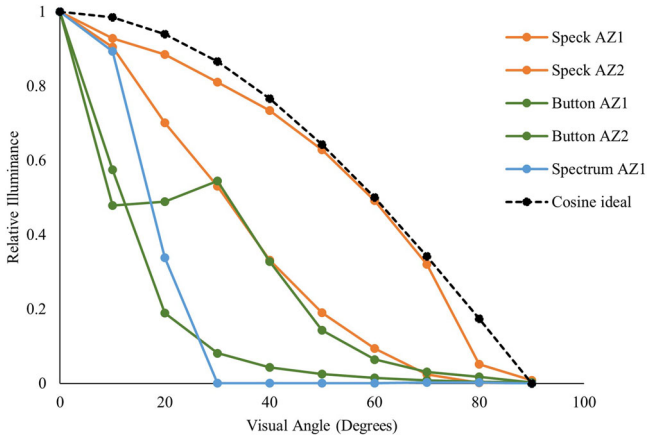


Fig. 7 | Directional response of the light dosimeters. The relative illuminance output of the Blue Iris Specks (orange lines), LYS Buttons (green lines), and Actiwatch Spectrum Plus (blue lines) dosimeters for various incident angles against a cosine-response reference (black dashed line).

Table 3 | f_2 indices for each light dosimeter and their corresponding classes for each device

Metric	Button	Speck	Actiwatch
f_2 index	9.33%	6.15%	11.98%
f_2 class	D	C	D

Directional response index (f_2)

Measurements were taken across a range of incident angles in two azimuthal planes, except for the Actiwatch, which recorded data for only one. Figure 7 presents the relative illuminance response of each dosimeter normalized to the 0° measurement and plotted against angle of incidence, alongside the theoretical cosine curve. A cosine response is the ideal target for illuminance meters and is shown here as a reference benchmark for angular responsivity. For wearable dosimeters, the most appropriate angular response can depend on the intent of use (e.g., approximating eye-level exposure and the wearer's effective field of view), so the cosine curve is used primarily for comparison rather than as a universal target.

Among the devices tested, the Speck dosimeters demonstrated the closest alignment with the cosine curve across both azimuth planes (Speck AZ1 and Speck AZ2), particularly in the 0°–60° range, showing a gradual decrease in response. Button dosimeters deviated more sharply from the theoretical cosine curve, with relative illuminance dropping significantly after 10°, and for another azimuth plane, relative illuminance plateaued below 0.2 lx for angles beyond 30°. Actiwatch dosimeter exhibited the poorest performance, with a sharp decline to near-zero response by 30°, indicating a highly directional and non-cosine behavior.

The calculated f_2 indices for each device, averaged across azimuth planes, are reported in Table 3 Speck achieved the lowest f_2 value at 6.15%, corresponding to Class C performance. The Button dosimeters had a higher f_2 of 9.33%, while the Actiwatch recorded the highest f_2 index at 11.98%, placing both devices in Class D. These classifications are based on DIN 5032-7 standards[97], where Class A and B correspond to more accurate cosine conformity (<3% and < 6%, respectively), and Class D indicates poor directional response (>6%). These findings underscore significant differences in angular accuracy between dosimeter types, with the Speck device demonstrating comparatively better performance, yet still not meeting Class B criteria.

Temperature dependence function ($f_{6,T}$ (T))

The temperature dependence function $f_{6,T}$ (T) has been calculated across a wide range of temperatures from −4.5 °C to 35 °C. However, it is important

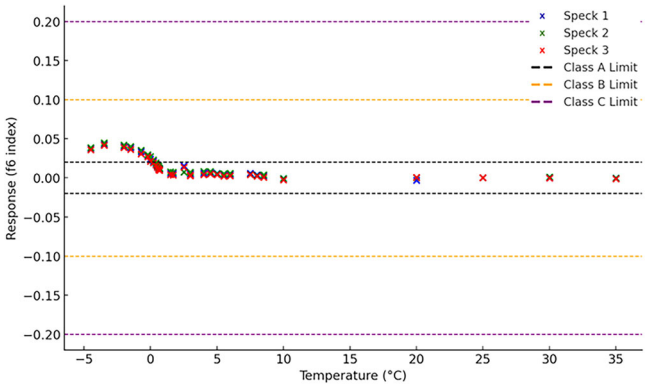


Fig. 8 | Temperature dependence of Blue Iris Speck dosimeters. The temperature dependence function $f_{6,T}$ (T) calculated for three Blue Iris Specks (blue, green, and red asterisks) and class limits (black, purple, and yellow lines) to provide visual context for the magnitude of the observed $f_{6,T}$ (T) deviations across temperature.

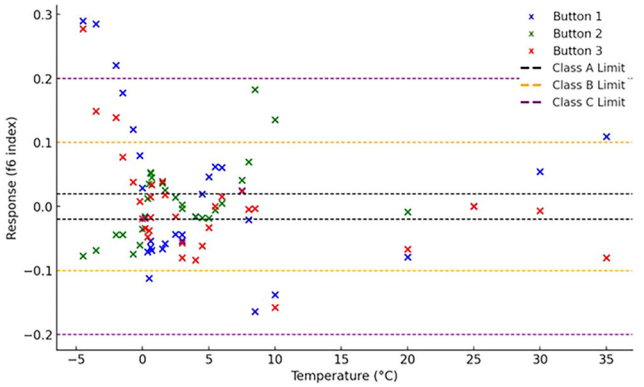


Fig. 9 | Temperature dependence of LYS Button dosimeters. The temperature dependence function $f_{6,T}$ (T) calculated for three LYS Buttons (blue, green, and red asterisks) and class limits (black, purple, and yellow lines) to provide visual context for the magnitude of the observed $f_{6,T}$ (T) deviations across temperature.

to consider the practical use of these devices when worn on the body, where temperature exposure is generally less variable. Indoors, for example, the device is subject to ambient temperatures that can vary substantially depending on the local climate and building operation[98] with some potential influence of skin temperature if the device is in direct contact. In a typical indoor environment (~23 °C), mean (weighted) skin temperature in resting adults is often reported in the low-to-mid 30 s in °C, though local skin temperature varies substantially by body region[99–101]. Therefore, for indoor applications, a narrower temperature range between 20 °C and 35 °C might suffice to assess device performance, aligning closer to actual operating conditions.

For each device type, dashed lines representing the tolerance levels (Class A, Class B, and Class C) serve as reference points (Figs. 8–10), allowing us to assess whether the responses remain within acceptable limits as temperatures change. Because ISO/CIE 19476 defines this tolerance for the single-number temperature-dependence quality index $f_{6,T}$ (not the temperature-dependent function $f_{6,T}$ (T)), the class lines shown here are only to provide visual context for the magnitude of the observed $f_{6,T}$ (T) deviations across temperature. Observing the scatter across the plots, the extent of variability can be assessed within each device type, and the temperature responses between different devices may be compared.

Among the Speck devices, the responses remained relatively close to zero across temperatures Fig. 8, with only minor deviations in values that seem to stabilize around the baseline. Similarly, the Button devices showed more fluctuations, especially at lower temperatures, although their

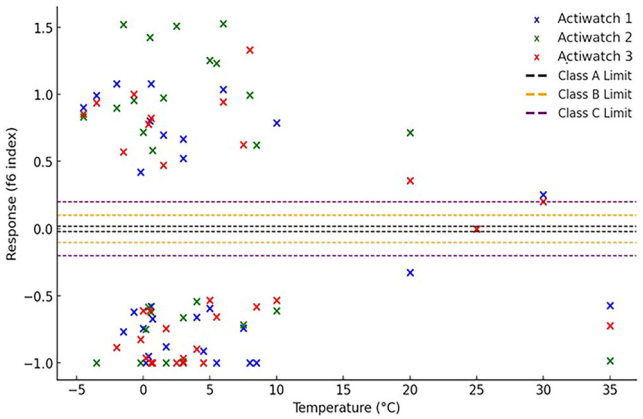


Fig. 10 | Temperature dependence of Actiwatch Spectrum Plus dosimeters. The temperature dependence function $f_{6,T}(T)$ calculated for three Actiwatch Spectrum Plus (blue, green, and red asterisks) and class limits (black, purple, and yellow lines) to provide visual context for the magnitude of the observed $f_{6,T}(T)$ deviations across temperature.

responses generally converged around zero as the temperature increased (Fig. 9). Conversely, the Actiwatch dosimeters showed larger variances, with responses peaking both positively and negatively across the temperature range, suggesting greater sensitivity to temperature changes compared to Speck and Button devices.

To quantify these observations, an analysis of variance (ANOVA) was conducted both within each device type and between the three device types. The within-device analysis examined the three models within each device type to assess the consistency of temperature dependence among them. The ANOVA results indicated variations within the Actiwatch dosimeters ($p < 0.05$), which had the most pronounced fluctuations in response, while the Speck ($p = 0.086$) and Button dosimeters ($p = 0.134$) exhibited non-significant differences, indicating more consistency within each model.

For the between-device analysis, average responses for each temperature across the three types were compared (i.e., Speck, Button, and Spectrum). This ANOVA showed differences ($p < 0.05$) between the device types, confirming that the temperature dependence varies among them, with Actiwatch dosimeters being the most temperature sensitive. These findings underscore the importance of considering temperature effects when using these devices in varying environmental conditions. In practical terms, the results suggest that Actiwatch dosimeters may require additional calibration or compensation mechanisms for precise measurements across diverse temperatures, while Speck and Button devices demonstrate greater stability with less sensitivity to temperature fluctuations.

Finally, three indices were calculated for each portable device, highlighting their potential inaccuracies. This study underscores the importance of understanding the true meaning of the output values provided by these devices. Too often, it is assumed that the values displayed by the devices are accurate. However, the findings here clearly demonstrate that this assumption may be incorrect.

Discussion

Melanopic light measurements require accurate determination of the amount of light or the direction of light incidence. Since the position of wearable dosimeters (e.g., wrist) may deviate significantly from the eye position, the circadian light measurement can greatly diverge from realistic values. In this study, three commercially available wearable light dosimeters (LYS Button, Blue Iris Speck, and Actiwatch Spectrum Plus) were benchmarked against a calibrated spectroradiometer under controlled LED (0 lx–570 lx) and partly cloudy daylight conditions (42,000 lx–78,000 lx) and were further evaluated for directional response (f_2) and temperature dependence ($f_{6,T}(T)$).

Button and Speck performed similarly to the reference spectroradiometer, but Actiwatch Spectrum Plus had the poorest performance under both LED and daylight exposure, with significant differences among the three Actiwatch dosimeters. LYS Button and Blue Iris Speck dosimeters also performed poorly under daylight. From the results of this study and the pilot study[94], it was observed that all the available wearable light dosimeters had poor performance under higher light levels, especially under daylight (>20,000 lx). Thermal effects due to solar gain are discussed separately in the temperature dependence discussion below. A more detailed study looking at the illuminance measurement accuracy threshold under a fixed light source for each dosimeter should be performed.

The comparison between Blue Iris Speck and LYS Button dosimeters illustrates that device selection affects both measurement performance and the feasibility of circadian light measurement in practice. While both align with the reference under high illuminance (>200 lx), the LYS Button's consistent overestimation at low-light levels, by up to 1 lx at a reference mel-EDI of 5.9 lx, suggests it may inflate estimates of biologically meaningful light in dim settings. These findings highlight the importance of evaluating dosimeter behavior across the full dynamic range and suggest that the Speck may offer more reliable performance for studies requiring sensitivity to circadian stimuli. Considering the recent suggestions by the CIE that vertical mel-EDI at the eye-level should be limited to 10 lx 3 h before bedtime and to a minimum during sleep[102], the dosimeter accuracy in low-light levels is important for sleep research[38].

The observed spectral deviations between the Blue Iris Speck dosimeter and the reference spectroradiometer likely reflect inherent limitations in the dosimeter's hardware architecture. Although the Blue Iris Speck effectively captured the general shape of the daylight spectrum, discrepancies in both amplitude and peak structure, particularly under LED illumination, suggest limited spectral reconstruction fidelity and source-dependent reconstruction artifacts. This may stem from a small number of channels or broader spectral bins in the Blue Iris Speck compared to the high-resolution photodiode arrays or charge-coupled device (CCD) detectors used in research-grade reference instruments. Figures 4 and 5 illustrate that reconstruction performance is source dependent. The broad daylight spectrum is approximated more successfully than the narrowband structure of PC white LEDs. The elevated standard error observed in specific wavelength ranges, especially beyond 600 nm, could indicate undersampling or the absence of dedicated photo detectors for those bands. Such limitations are known trade-offs in miniaturized dosimeters, where size and power constraints restrict the number of channels that can be accommodated. While these design choices enable portability and wearability, they compromise spectral fidelity, particularly when measuring complex light sources, such as LEDs with narrowband spectral distributions. A limitation is that the LED spectral comparisons were performed under controlled, source-based laboratory conditions, whereas the daylight spectral comparisons were inherently scene-based (i.e., conducted outdoors) and may cause geometry, reflection, and orientation-driven variability beyond spectral performance alone. This highlights the importance of understanding dosimeter limitations when interpreting spectral data from compact dosimeters during in-field applications.

It is worth noting that some of the observed differences in directional response may be influenced by the physical design of each dosimeter. The Blue Iris Speck dosimeter, which exhibited the most cosine-like angular response, as shown in the results above (Fig. 7), features a smoothly curved and diffused lens design that likely facilitates more uniform light collection from a wide range of angles. In contrast, the LYS Button's narrow V-shaped optical windows may restrict off-axis light entry, contributing to its steeper signal drop-off. The Actiwatch dosimeter houses its photodiode within a recessed optical well, which is expected to significantly limit angular acceptance and explain its rapid decline in sensitivity below 30° incidence angle. This behavior is consistent with prior laboratory characterization of the Actiwatch Spectrum Plus dosimeters showing large cosine-response (high f_2) errors attributed to optical geometry effects[70]. This interpretation, based on visual inspection of device geometry, suggests that optical housing and lens design play a critical role in determining angular performance. Further optical modeling and measurements would be required to quantify

the contribution of these geometric features to the observed f_2 differences and overall angular responsivity.

In general, a low f_2 index indicates closer cosine conformity and therefore better angular accuracy for conventional illuminance meters, since the measured signal is expected to vary with the cosine of the angle of incidence. For wearable dosimeters used in chronobiology research, however, the interpretation is more nuanced. The biologically relevant exposure is ocular rather than simply planar incident light, and prior work has shown that dosimeter location on the body can substantially affect agreement with eye-exposure[30,31], with wrist-worn measurements being less representative than location nearer the head or corneal plane. Therefore, the cosine response quality index remains a useful standardized descriptor of angular behavior and a valid basis for comparing devices, but it should not be interpreted as a standalone indicator of circadian-measurement validity. In that context, the cosine curve is best treated as a reference benchmark for inter-device comparison, while the practical relevance of a given angular response should be considered in relation to the intended use of the dosimeter, such as approximating eye-level exposure in real-world settings[31].

This study shows that temperature significantly affects the accuracy of light measurement devices, especially Spectrum model dosimeters, which exhibit high sensitivity to temperature changes. Speck devices are more stable across temperatures, while Button and Actiwatch dosimeters may require extra calibration to ensure accuracy. The oscillating behavior visible in the temperature-dependence curves for Button devices compared to the Speck devices, which did not demonstrate the same behavior, explains possible coarse or internal correction within the device and is less likely to originate from the setup itself. While present data does not allow for analyzing the underlying mechanism, this oscillating behavior suggests an additional source of uncertainty within Button devices when used under varying thermal conditions. These findings highlight the importance of considering temperature effects for reliable measurements, because output values should not always be assumed to be accurate without temperature compensation. In field deployments, thermal exposure may not be determined by ambient air temperature alone: when worn in direct sun, the device housing can experience solar loading (case heating) and reach temperatures above ambient. This can amplify temperature-related measurement errors.

The wearable light dosimeters are designed with the goal of recording personal light exposure in field conditions. Consequently, many factors other than photometric accuracy play vital roles in the usability of these wearable light dosimeters, such as sufficient battery life, accurate data-logging, user-friendly interface, ease of setting up or pairing the dosimeter, and inobtrusive wearability as close to eye-level as possible.

Based on the ease of use, LYS Buttons can be ranked as the most user-friendly, with the caveat of compromised photometric accuracy within the available light dosimeters. Although the LYS Button dosimeter is easy to pair with its smartphone application, it has shown data-upload issues with Android smartphones in the pilot study, and thus only iOS smartphones were used in this study. The following are precautions recommended for LYS Buttons:

- They are water- and dust-resistant (IP54) but should not be immersed in water.
- The average battery life was observed to be 5 days, and devices should be charged accordingly.
- Data gets overwritten every 18 h, and therefore, the device should be paired with a smartphone with an active internet connection at least once within that timeframe to upload the data to the cloud.
- All the uploaded data are available to download through the subscribed researcher account; when performing a longitudinal study, it is recommended that the data be backed up once every week.
- The timestamps of the logged data for each dosimeter may not match the local time zone; consequently, manually recording the clock time once the device starts uploading the data to the cloud is recommended.
- Users should be aware that clothing layers or hair may cover the dosimeter and cause inaccuracies in the data.

Blue Iris Specks had the highest photometric accuracy, but they were also the most difficult to pair with the gateway. While the newer Speck version (v2.0) is released with Bluetooth pairing, onboard memory of 180 days, and one year of battery life, the version of Speck (v1.0) that was available for this study had no onboard memory, non-rechargeable 3 V lithium batteries, no Bluetooth pairing option, and only connected to the cloud through a gateway. Although the manufacturer indicated that Speck continues logging at 1-min interval when offline (i.e., without an active internet connection) and stores data in internal flash until uploaded to the cloud, data loss is expected only if the battery is depleted. However, during this study, we observed missing data samples, suggesting that pairing and/or connectivity instability can still result in incomplete datasets under field conditions. The following are precautions recommended for Blue Iris Specks (v1.0):

- The dosimeters must be within a 7 m radius of the gateway to avoid data loss.
- The gateway needs an active, uninterrupted internet connection for the entire period of data collection; weak internet connections might cause loss of pairing and thus loss of data.
- A red LED indicates that it is turned on, reset, and ready to pair with a gateway. A short blue-white flicker LED indicates that it is successfully connected to a gateway and is uploading the data to the cloud.
- Users should be aware that clothing layers or hair may cover the dosimeter and cause inaccuracies in the data.

The operation of Actiwatch Spectrum Plus is simple. To extract recorded data from onboard storage, the device must be hardwired to a computer via USB, and the manufacturer's software (Philips Actiware v6.0.4) is used to download the data into a CSV file format. The Actiwatch dosimeter is robust and water-resistant. However, Actiwatch Spectrum and Spectrum Plus have been discontinued and are no longer manufactured or supported; the points below are intended for groups using existing devices and for interpreting legacy datasets. The following are precautions:

- Calibrate the device before deploying it in the field.
- Due to the wrist-worn design and poor photometric performance, the light exposure data are not reliable[31]; in a research setting, instead of using it to measure light exposure, it can be used to validate light versus no light conditions or validate light data obtained from other wearable light dosimeters on the same user.
- It does not provide any melanopic metrics (mel-EDI or CS) but does provide irradiance (unit: $\mu W/cm^2$) –irradiance reported from the dosimeter has short wavelength sensitivity that is not identical to melanopic quantities and should not be used interchangeably.

As for the method described for investigating the f_2 index, the following precautions should be taken before measuring:

- Ensure a completely uniform illuminance distribution on the measurement surface caused by the test light source.
- Check whether the center of the light source has its focus exactly at the same point on the table at varying angles of incidence.
- Check the aimed angles between the source and the dosimeter.
- Ensure that the distances between the light source and the dosimeters remain constant for the different angles.

As for the method described for investigating the f_6 index, the following precautions should be taken before measuring:

- Stable reference and accurate temperature control: Ensure a stable reference temperature (typically 25 °C) and precise control at each

temperature point to avoid fluctuations that could affect the f_6 index measurements.
- Allow for thermal equilibration: Let the device and surroundings reach thermal equilibrium at each temperature setting before taking measurements to prevent inaccuracies from transient thermal effects.
- Minimize environmental interference: Keep external conditions (e.g., humidity, airflow, and light) constant during measurements, because changes in these factors can impact the device's response.

Limitations of this work include the small sample size (three units per device type), a short daylight sampling period under single sky conditions at a single location, LED testing outside an integrating sphere, spectral comparison based on a single Blue Iris Speck unit, and the absence of body-worn placement and occlusion effects. Further studies should replicate these tests across a larger number of devices and updated hardware versions, quantify saturation thresholds under controlled high illuminance, and evaluate eye-level versus wrist or chest wearing positions. Apply device-specific field calibration and temperature-compensation approaches to support reliable use in longitudinal field studies[65]. Wearable light dosimeters are increasingly being used to quantify personal light exposure for circadian, sleep, mood, and alertness research, but their outputs should not be assumed to be interchangeable without device-specific characterization[31,103]. Taken together, the Blue Iris Speck offered the strongest overall measurement fidelity for illuminance, mel-EDI, and angular response. The LYS Button provided the simplest field deployment workflow but showed reduced accuracy in daylight and at low-light mel-EDI. The Actiwatch Spectrum remains valuable for actigraphy, but, given its photometric variability and lack of melanopic outputs, should be used cautiously for quantitative light-dose interference in studies where light dose is a primary exposure variable[78].

Methods

In this study, three wearable light dosimeters were evaluated using a structured protocol designed to quantify photometric performance, melanopic metrics, spectral behavior, directional response, and temperature dependence under repeatable conditions. Because no single widely adopted standard exists for on-body light dosimeters evaluation, relevant performance indices from ISO/CIE 19476:2014 (developed for illuminance meters) were used as a comparative framework, especially for directional response and thermal sensitivity, while realizing real-world wear introduces additional variability (e.g., body placement, motion, and partial occlusion) not captured by these indices. All testing was conducted at the Pennsylvania State University on the University Park campus, Pennsylvania, USA. The indoor measurements were performed in the neutrally painted (Munsell N8) lighting lab. A 15-channel multi-color LED lighting system (LEDCube by Thouslite) and MATLAB® were used to generate the light stimuli representing LED light sources.

Dosimeters tools

Three different types of wearable light dosimeters were included in this study, as shown in Fig. 1. All three dosimeters selected for this study were commercially available at the time this study began (September 2023). The current study was conceptually initiated in 2022, at the end of the COVID-19 pandemic. Therefore, the selection of dosimeters was primarily influenced by their commercial availability and widespread use in research. Additionally, other devices identified in the literature were either unavailable for loan or purchase due to shipping delays, global microchip shortages during the COVID-19 pandemic, insurance requirements, or budgetary constraints. Manufacturing of the Actiwatch Spectrum and Actiwatch Spectrum Plus by Phillips was discontinued in January 2023.

The Actiwatch Spectrum Plus can be used for continuously measuring activity and light exposure. Its RGB photosensor is located within a recessed optical well Fig. 1, which constrains off-axis light entry and thus influences angular responsivity. As the name implies, the Actiwatch is designed to be worn on the wrist. The technical specifications of the activity and light measuring system in the Actiwatch are not publicly available; however, the manufacturer did report a spectral sensitivity range between 400 and 700 nm for its RGB photosensor system, and the spectral sensitivity of the older versions was previously reported[64,67]. The manufacturer reported an accuracy of 10% at 1500 lux when measuring photopic illuminance. A report on the photometric characteristics of the RGB photosensor system of the Actiwatch Spectrum has been published[67]. To collect the personal light exposure data, participants can be sent out in the field wearing an Actiwatch for up to 60 days (maximum battery life), and then these data can be downloaded to a Windows® computer by connecting the Actiwatch via a USB cable. Actiwatch calculates white light illuminance by integrating the input from separate red, green, and blue channels. It measures daily total light exposure (unit: lx-min), average illuminance (unit: lx), and irradiance ($\mu W/cm^2$) for all channels every 30 s. Although Actiwatch has onboard data storage for 60 days, this stored data cannot be retrieved remotely, and the device must be returned to the researcher to download the data.

Button and Speck dosimeters upload the measured data directly to the cloud, making them more suitable for longitudinal studies. The LYS button dosimeter is manufactured by LYS Technologies. Button's input optics consist of narrow V-shaped optical windows (Fig. 1), which can reduce off-axis light collection relative to a diffused aperture. The LYS button dosimeter connects to the LYS Track or LYS Collect application in a smartphone via Bluetooth; the smartphone then uploads the data to the cloud, and these logged data are accessible through an online dashboard. Buttons outputs are illuminance, melanopic lux, red, green, and blue (RGB) diode irradiance values, infrared (IR) values, correlated color temperature (CCT)[104], and it records movement using an accelerometer, once every 15 s. The LYS button dosimeter must connect to the internet via smartphone at least every 18 h to ensure complete data collection. After this period, the LYS button dosimeter will begin to overwrite previously stored data to create room for the next cycle.

The greatest advantage of the Speck is its ability to output a reconstructed estimate SPD reported at 5 nm increments. The Speck uses a smoothly curved, diffused input lens (Fig. 1), which supports broader angular acceptance. Personal communication with the Blue Iris manufacturer confirmed that the dosimeter utilizes seven channels to measure the SPD. Blue Iris Speck records illuminance (unit: lx), CCT (unit: K), CS, mel-EDI (unit: lx), the CIE 1931 (x, y) chromaticity coordinates[105], and near infrared (NIR) value once every minute. Although communication with the manufacturer confirmed that the upcoming Speck (v2.0) will require Bluetooth connectivity to transfer the data, the tested version of the Speck (v1.0) needed an uninterrupted connection with a gateway to ensure complete data transfer to the cloud. To begin performing measurements with Specks, the gateway first needs to be connected to an internet network using a smartphone or computer, after which the Speck must be reset using a pinhole button. Once the Speck devices are synced, real-time illuminance measurements can be seen on the research data portal. These steps required extensive trial-and-error, complicating data collection for the available version (v1.0) of the Blue Iris Specks. Table 4 shows the overall comparison of these three wearable devices and detailed technical information of the investigated dosimeters.

Setup

A calibrated Gigahertz-Optik BTS256 spectroradiometer was used as the reference measurement device. Data was collected in two phases. In the first phase, the experiment was set up to evaluate the agreement of the collected data using dosimeters with a reference instrument under different light sources (LED, incandescent, and daylight) and at different light levels. In the second phase, the experiment was designed to evaluate the accuracy of wearable light dosimeter data collection across different incident angles. In the third phase, the experiment was designed to evaluate the temperature dependence of the dosimeter measurements, i.e., the effect of ambient temperature on measurement accuracy.

In this manuscript, "accuracy" refers to agreement with the reference instrument (systematic bias). The "spectral accuracy" refers to how close the

Table 4 | Technical characteristics of the three wearable light dosimeters

Device	Mounting	Dimensions & weight	Battery & runtime	Spectral range	Other features	Connectivity & s.oftware
LYS Button	Clip	Ø: 24 mm h: 12 mm, 4.0 g	Rechargeable, 7 days	R, G, B, IR (380–1100 nm)	Mel-EDI measurement, light exposure, 3-axis accelerometer, IP54, Sampling rate 15 s	BLE, Cloud access, Software: LYS Track, LYS Collect (Android, iOS)
Blue Iris Speck	Pendant, Clip	Ø: 29 mm h: 15 mm, 10 g	None, 1 year	Reconstructed SPD estimate (350–770 nm, 5 nm steps)	Calculation of illuminance, CCT, EML, CS, and C-Rl. Temp & humidity measurement	BLE, Cloud access, Software: Speck Connect
Actiwatch Spectrum Plus	Wrist band	48 × 37 × 15 mm, 31 g	Rechargeable (USB), 60 days	R, G, B (350–700 nm)	Event marker, visual feedback, audible off-wrist reminder, alarm reminder, uniaxial accelerometer	USB, Software: Actiware (Windows)

device's spectral measurement was to the reference as a function of wavelength and "spectral range" refers to wavelength interval over which device provides a valid measurement. Random variability was reported separately as between-unit variability within each device type and as the spread of measurements across conditions; strict within-unit repeatability at fixed conditions was not strictly quantified.

The data from the reference meter was collected using desktop software (S-BTS256). Button dosimeters data were collected via the LYS smartphone application, allowing direct transfer to a spreadsheet for analysis. Speck data were recorded on the Blue Iris website, permitting real-time measurement tracking and downloadable data, which were additionally transferred to a spreadsheet. Light data collected by the Philips-Respironics Actiwatch Spectrum Plus were downloaded from the device using the software Philips Actiware version 6.0.4.

Data were consolidated into a single spreadsheet, after which samples of the lighting conditions were selected and compared to identify deviations from the control dosimeter and within the same dosimeter type. These comparisons were performed under various conditions, including LED light sources, daylight, and directionality measurements. Directional measurements were executed with the utilization of a heliodon.

Performance under LED

The reference calibrated spectroradiometer, three LYS Buttons, three Blue Iris Specks, and three Actiwatch Spectrums were placed adjacent on a tabletop at 0.76 m from the floor and exposed to an LED light source with controlled CCT (4600 K). Indoor light exposure measurements were conducted using the pcLED white channel of the multi-color tunable LED-Cubes (Thouslite) at 4600 K. Illuminance levels were adjusted from 100% (up to 570 lx) to 0% incrementally using MATLAB. The circadian applications are often concerned with low-light exposure, recommended mel-EDI thresholds on the order of 1 and 10 lx have been discussed in the literature[38,44]. Accordingly, the indoor LED protocol included targeted data points in the low-light range (including several measurements below 10 lx). All recorded data were stored on the cloud for LYS Buttons and Blue Iris Specks and on board for Actiwatch and retrieved later. The reference spectroradiometer was controlled remotely using desktop software via a data cable, as shown in Fig. 11.

Performance under daylight

Daylight measurements were conducted in partly cloudy sky conditions (42,000 lx–78,000 lx) around approximately 17:00 GMT-4 in State College, PA, on September 23rd, 2023. An open field, free of trees or tall buildings, was chosen for unobstructed measurements. The measurement platform was mounted on a tripod and leveled using a bubble level to maintain a horizontal measurement plane; the reference calibrated spectroradiometer and all dosimeters were mounted adjacent on the same rigid platform with their sensing apertures co-planar and fixed in position throughout the trial. Hence, all devices experienced the same incident conditions due to the shared leveled platform. Daylight measurements were conducted for roughly 20 min to assess the variability in daylight exposure. In addition, the short measurement window minimized changes in solar elevation. Because sunlight is inconsistent, measurements followed no clear pattern, and data recording was completed arbitrarily after sufficient data were collected to draw comparisons.

Directional response (f_2 index)

The index describing the responsivity of the photometer to light incident at an angle other than normal (relative to the cosine-law reference used for planar illuminance meters) is called f_2 index[106]. The directional response index (evaluation of the incident light as a function of the angle of incidence) is determined by the form and the optical construction of the photometer head. Thus, special evaluation functions can be realized by equipping the photometer head with directionally selective optical elements (e.g., diffusing adaptors of various shapes and special optical components). For a photometer with a plane input window measuring planar illuminances, the

deviation in directional response to the incident radiation is given by:

$$f_2(\varepsilon, \varphi) = \frac{R(\varepsilon, \varphi)}{R(0, \varphi)\cos(\varepsilon)} - 1 \quad (3)$$

where,

$R(\varepsilon,\varphi)$ is the output signal value as a function of the angle of incidence ε and azimuth angle φ;

ε is the incident angle measured with respect to the normal to the measuring plane or optical axis;

φ is the angle of rotation (azimuth angle) of the light sensor around the vertical axis.

For characterizing the directional response, $f_2(\varepsilon,\varphi)$ is measured in four orthogonal planes of azimuth φ = {0, π/2, π, 3 π/2}. For a given orientation the index f_2 is calculated as:

$$f_2(\varphi) = \int_0^{80^\circ} f_2(\varepsilon, \varphi)\ \sin(2\varepsilon)\, d\varepsilon \quad (4)$$

A cosine response is the conventional reference for planar illuminance meters[106] and provides a standardized comparison baseline for reporting f_2. With the available illuminance values per device, it was possible to calculate the f_2 index per angle of incidence by using Eq. (3), the final f_2 index of the dosimeter by Eq. (4), and a corresponding index class of each dosimeter in two azimuth planes was determined.

To measure the directional response, a directional LED light source (Lumenpulse, 3500 K, 28 W, 2381 lm) was set at a fixed distance from the photometric plane of the dosimeters. Next, dosimeters were mounted on a heliodon and aligned with the center of the light source. Alignment was performed by positioning the sensing aperture of each device on the heliodon's rotation axis and orienting the dosimeter measurement plane normal to the source at 0°. The setup was programmed to provide different angles of incidence on the dosimeters, as shown in Fig. 12. The dosimeters moved from 0° to 90° (aligned to the center of the light source) with 10° intervals every five minutes. Measurements were performed for two different azimuth planes (AZ1 and AZ2). For AZ1, the sensor was tilted from 0° to 90° while keeping the light source stationary. For AZ2, the dosimeter was first rotated by 90° about its own surface normal (i.e., around the axis perpendicular to the sensing plane), and the 0° to 90° tilt sequence was then repeated. During the measurements, the light source output and room temperature remained constant.

To reduce bias from stray/reflected light at other angles of incidence, a directional luminaire was used, and illuminance at the dosimeter plane was verified with the reference meter at 0° condition for the mounting position used, confirming comparable incident illuminance across placements. As the sensors were rotated while the light source remained stationary, the source -to-sensor distance was verified and corrected as needed after each 10° increment to maintain the intended geometry. A limitation is that device linearity in the narrow illuminance range immediately below the 0° condition (used for normalization of relative angular response) was not independently characterized within the directional-response setup; however, the 0° illuminance was selected to fall within the operational spectral range evaluated in the photometric comparisons. An additional limitation is that directional response was evaluated using photopic illuminance; melanopic-weighted angular response may differ due to wavelength-dependent responsivity and sensor/optics geometry, and should be quantified in future work where mel-EDI outputs are available.

Temperature dependence function ($f_{6,T}(T)$)

Light measurement devices and dosimeters in general can be sensitive to temperature variations, depending on the design and materials used, but this



Fig. 11 | Experimental setup for photometric measurements. Experimental setup for photometric measurements under LEDs (left) and daylight (right).



Fig. 12 | Experimental setup for directional-response measurements. Experimental setup to measure the directional response of dosimeters.

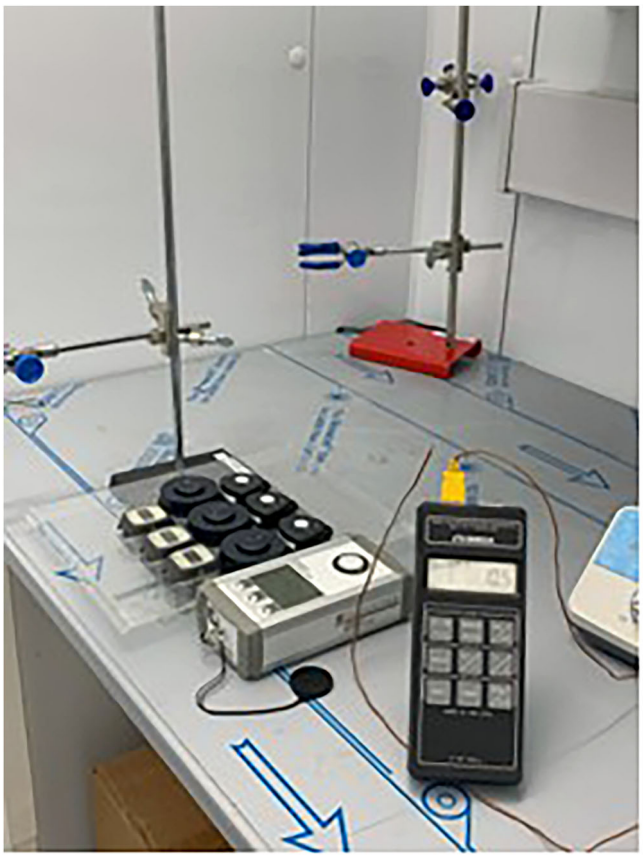


Fig. 13 | Experimental setup for temperature-dependence measurements. Experimental setup for temperature dependence function measurements.

has not been systematically investigated. Therefore, making generalizable measurements requires establishing that the sensor response is independent of temperature over the anticipated range of use. Temperature dependence characterizes the influence of the ambient temperature on the absolute responsivity and the relative spectral responsivity of the photometer[107]. If the photometer is operated at an ambient temperature different from that used during calibration, measurement errors can occur.

ISO/CIE 19476 defines the temperature-dependence characterization function $f_{6,T}(T)$ and a single-number quantity index $f_{6,T}$ computed at 40 °C. Because measurements at 40 °C were not performed in this study, we reported only $f_{6,T}(T)$ over the tested temperature range of −4.5 °C to 35 °C. The temperature range between 10 °C and 20 °C was not sampled because of a limitation of the climate-chamber control system available for this study, which did not permit stable operation or data collection across this interval.

To determine the temperature-dependence characterization $f_{6,T}(T)$, several preparatory steps are essential. First, the stability of the light source should be verified multiple times throughout the measurements to ensure that any observed variations are due to temperature changes and not fluctuations in light intensity. Second, the accuracy of temperature control within the climate room must be taken into account, because even slight deviations could impact the function calculation. These preparations are necessary to obtain reliable $f_{6,T}(T)$ values that genuinely reflect the temperature response of the devices.

All three dosimeters were exposed to different temperatures at a stable light level of 880 lx using the ceiling installed LEDs (CRI 80, 4000 K) in a climate chamber, as shown in Fig. 13. While the temperature of the chamber was varied between −4.5 °C and 35 °C using the wall-mounted control panel, the dosimeters were set to log the illuminance values simultaneously. The temperature values were verified and recorded using a digital thermometer (Omega HH24). The dosimeters were allowed to attain thermal equilibrium before starting the measurement.

The characterization of temperature dependence at a given ambient temperature (T) is given by the function $f_{6,T}(T)$:

$$f_{6,T}(T) = \frac{Y(T)}{Y(T_0)} - 1 \quad (5)$$

where,

$Y(T)$ is the output signal at temperature T,

$Y(T_0)$ is the output signal at 25 °C reference ambient temperature.

Human subjects research

This study did not involve human participants, human biological materials, or identifiable human data. Therefore, ethics committee/institutional review board approval and informed consent were not required.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available because they require additional de-identification and accompanying documentation to support responsible reuse and prevent misinterpretation, but are available from the corresponding author on reasonable request.

Received: 18 July 2025; Accepted: 20 May 2026;
Published online: 21 July 2026

References

1. Schlangen, L. J. M. & Price, L. L. A. The lighting environment, its metrology, and non-visual responses. *Front. Neurol.* **12**, 624861 (2021).
2. Boivin, D. B., Duffy, J. F., Kronauer, R. E. & Czeisler, C. A. Dose-response relationships for resetting of human circadian clock by light. *Nature* **379**, 540–542 (1996).
3. Khalsa, S. B. S., Jewett, M. E., Cajochen, C. & Czeisler, C. A. A phase response curve to single bright light pulses in human subjects. *J. Physiol.* **549**, 945–952 (2003).
4. Lewy, A. J. Effects of light on human melatonin production and the human circadian system. *Prog. Neuropsychopharmacol. Biol. Psychiatry* **7**, 551–556 (1983).
5. St Hilaire, M. A. et al. The spectral sensitivity of human circadian phase resetting and melatonin suppression to light changes dynamically with light duration. *Proc. Natl. Acad. Sci. USA* **119**, e2205301119 (2022).
6. Gubin, D. et al. Blue light and temperature actigraphy measures predicting metabolic health are linked to melatonin receptor polymorphism. *Biology* **13**, 22 (2023).
7. Ishihara, A., Courville, A. B. & Chen, K. Y. The complex effects of light on metabolism in humans. *Nutrients* **15**, 1391 (2023).
8. Cajochen, C., Zeitzer, J. M., Czeisler, C. A. & Dijk, D.-J. Dose-response relationship for light intensity and ocular and electroencephalographic correlates of human alertness. *Behav. Brain Res.* **115**, 75–83 (2000).
9. Chellappa, S. L. et al. Non-visual effects of light on melatonin, alertness and cognitive performance: can blue-enriched light keep us alert? *PLoS ONE* **6**, e16429 (2011).
10. Milosavljevic, N. How does light regulate mood and behavioral state? *Clocks Sleep.* **1**, 319–331 (2019).
11. Rahman, S. A. et al. Diurnal spectral sensitivity of the acute alerting effects of light. *Sleep* **37**, 271–281 (2014).
12. Vandewalle, G., Maquet, P. & Dijk, D.-J. Light as a modulator of cognitive brain function. *Trends Cogn. Sci.* **13**, 429–438 (2009).
13. Vandewalle, G. et al. Spectral quality of light modulates emotional brain responses in humans. *Proc. Natl. Acad. Sci. USA* **107**, 19549–19554 (2010).
14. Rosenthal, N. E. et al. Melatonin in seasonal affective disorder and phototherapy. *J. Neural Transm. Suppl.* **21**, 257–267 (1986).

15. Figueiro, M. G., Eggleston, G. & Rea, M. S. Effects of light exposure on behaviour patterns of Alzheimer's disease patients: a pilot study. In *Proc. Light and Human Health: EPRI/LRO 5th International Lighting Research Symposium 151–156. Electric Power Research Institute Lighting Research Office, Palo Alto, CA* (2002).
16. Van Someren, E. J. W., Kessler, A., Mirmiran, M. & Swaab, D. F. Indirect bright light improves circadian rest-activity rhythm disturbances in demented patients. *Biol. Psychiatry* **41**, 955–963 (1997).
17. Brandon, D. H., Holditch-Davis, D. & Belyea, M. Preterm infants born at less than 31 weeks' gestation have improved growth in cycled light compared with continuous near darkness. *J. Pediatr*. **140**, 192–199 (2002).
18. Rivkees, S. A., Mayes, L., Jacobs, H. & Gross, I. Rest-Activity Patterns of Premature Infants Are Regulated by Cycled Lighting. *Pediatrics* **113**, (2004).
19. Hattar, S., Liao, H.-W., Takao, M., Berson, D. M. & Yau, K.-W. Melanopsin-containing retinal ganglion cells: architecture, projections, and intrinsic photosensitivity. *Science* **295**, 1065–1070 (2002).
20. Berson, D. M., Dunn, F. A. & Takao, M. Phototransduction by retinal ganglion cells that set the circadian clock. *Science* **295**, 1070–1073 (2002.
21. Bailes, H. J. & Lucas, R. J. Human melanopsin forms a pigment maximally sensitive to blue light ($\lambda_{max} \approx 479$ nm) supporting activation of $G_{q/11}$ and $G_{i/o}$ signalling cascades. *Proc. R. Soc. B Biol. Sci.* **280**, 20122987 (2013).
22. CIE. CIE S 026/E:2018: CIE System for Metrology of Optical Radiation for ipRGC-Influenced Responses to Light. CIE Central Bureau, Vienna https://doi.org/10.25039/S026.2018 (2018).
23. Rea, M. & Figueiro, M. Light as a circadian stimulus for architectural lighting. *Light. Res. Technol.* **50**, 497–510 (2018).
24. Barroso, A., Simons, K. & de Jager, P. Metrics of circadian lighting for clinical investigations. *Light. Res. Technol.* **46**, 637–649 (2014).
25. Konis, K. A novel circadian daylight metric for building design and evaluation. *Build. Environ.* **113**, 22–38 (2017).
26. Durmus, D. Multi-Objective Optimization Trade-offs for color rendition, energy efficiency, and circadian metrics. in vol. 11706 117061J (International Society for Optics and Photonics, 2021).
27. Huang, Y., Li, J. & Dai, Q. Comparative analysis of circadian lighting models: melanopic illuminance vs. circadian stimulus. *Opt. Express* **32**, 29494 (2024).
28. Biller, A. M., Balakrishnan, P. & Spitschan, M. Behavioural determinants of physiologically-relevant light exposure. *Commun. Psychol.* **2**, 114 (2024).
29. Peeters, S. T., Smolders, K. C. H. J. & de Kort, Y. A. W. What you set is (not) what you get: how a light intervention in the field translates to personal light exposure. *Build. Environ.* **185**, 107288 (2020).
30. Stampfli, J. et al. The light-dosimeter: a new device to help advance research on the non-visual responses to light. *Light. Res. Technol.* **55**, 474–486 (2023).
31. Aarts, M. P. J., van Duijnhoven, J., Aries, M. B. C. & Rosemann, A. L. P. Performance of personally worn dosimeters to study non-image forming effects of light: assessment methods. *Build. Environ*. **117**, 60–72 (2017).
32. Hartmeyer, S., Webler, F. & Andersen, M. Towards a framework for light-dosimetry studies: Methodological considerations. *Light. Res. Technol.* **55**, 377–399 (2023).
33. Doulos, L., Tsangrassoulis, A. & Topalis, F. V. The role of spectral response of photosensors in daylight responsive systems. *Energy Build* **40**, 588–599 (2008).
34. Krüger, U., Ferrero, A., Thorseth, A., Mantela, V. & Sperling, A. General *V* (λ) mismatch index: History, current state and new ideas. *Light. Res. Technol.* **55**, 420–432 (2023).
35. Van Duijnhoven, J. et al. Measuring light exposure in daily life: a review of wearable light loggers. *Build. Environ.* **274**, 112771 (2025).
36. Zhao, X., Stafford, B. K., Godin, A. L., King, W. M. & Wong, K. Y. Photoresponse diversity among the five types of intrinsically photosensitive retinal ganglion cells. *J. Physiol.* **592**, 1619–1636 (2014).
37. Enezi, J. A. et al. A "melanopic" spectral efficiency function predicts the sensitivity of melanopsin photoreceptors to polychromatic lights. *J. Biol. Rhythms* **26**, 314–323 (2011).
38. Brown, T. M. et al. Recommendations for daytime, evening, and nighttime indoor light exposure to best support physiology, sleep, and wakefulness in healthy adults. *PLoS Biol.* **20**, e3001571 (2022).
39. CIE. CIE S 026/E:2018 CIE System for Metrology of Optical Radiation for ipRGC-Influenced Responses to Light. https://doi.org/10.25039/S026.2018 (2018).
40. Lucas, R. J., Lall, G. S., Allen, A. E. & Brown, T. M. How rod, cone, and melanopsin photoreceptors come together to enlighten the mammalian circadian clock. in *Progress in Brain Research* **199**, 1–18 (Elsevier, 2012).
41. Brown, T. M. Melanopic illuminance defines the magnitude of human circadian light responses under a wide range of conditions. *J. Pineal Res.* **69**, e12655 (2020).
42. Cajochen, C., Stefani, O., Schöllhorn, I., Lang, D. & Chellappa, S. Influence of evening light exposure on polysomnographically assessed night-time sleep: a systematic review with meta-analysis. *Light. Res. Technol.* **54**, 609–624 (2022).
43. Fazlali, F. et al. Lack of strong evidence for cone photoreceptors' contribution to human melatonin suppression and alerting response to light. *iScience* **28**, 112983 (2025).
44. Giménez, M. C. et al. Predicting melatonin suppression by light in humans: unifying photoreceptor-based equivalent daylight illuminances, spectral composition, timing and duration of light exposure. *J. Pineal Res.* **72**, e12786 (2022).
45. Hommes, V. & Giménez, M. C. A revision of existing Karolinska Sleepiness Scale responses to light: a melanopic perspective. *Chronobiol. Int.* **32**, 750–756 (2015).
46. Schöllhorn, I. et al. Melanopic irradiance defines the impact of evening display light on sleep latency, melatonin and alertness. *Commun. Biol.* **6**, 228 (2023).
47. Spitschan, M., Lazar, R., Yetik, E. & Cajochen, C. No evidence for an S cone contribution to acute neuroendocrine and alerting responses to light. *Curr. Biol.* **29**, R1297–R1298 (2019).
48. Papamichael, C., Skene, D. J. & Revell, V. L. Human nonvisual responses to simultaneous presentation of blue and red monochromatic light. *J. Biol. Rhythms* **27**, 70–78 (2012).
49. Paul, K. N., Saafir, T. B. & Tosini, G. The role of retinal photoreceptors in the regulation of circadian rhythms. *Rev. Endocr. Metab. Disord.* **10**, 271–278 (2009).
50. Spitschan, M. Melanopsin contributions to non-visual and visual function. *Curr. Opin. Behav. Sci.* **30**, 67–72 (2019).
51. Song, W. & Durmus, D. Evaluating energy efficiency and colorimetric quality of electric light sources using alternative spectral sensitivity functions. *Buildings* **12**, 2220 (2022).
52. Wyszecki, Günter. & Stiles, W. S. *Color Science: Concepts and Methods, Quantitative Data, and Formulae*, 1st ed. (*John Wiley Sons*, 2000).
53. Chang, A., Scheer, F. A. J. L. & Czeisler, C. A. The human circadian system adapts to prior photic history. *J. Physiol.* **589**, 1095–1102 (2011).
54. Dewan, K., Benloucif, S., Reid, K., Wolfe, L. F. & Zee, P. C. Light-induced changes of the circadian clock of humans: increasing duration is more effective than increasing light intensity. *Sleep* **34**, 593–599 (2011).
55. Foster, R. Fundamentals of circadian entrainment by light. *Light. Res. Technol.* **53**, 377–393 (2021).

56. Lucas, R. J. et al. Measuring and using light in the melanopsin age. *Trends Neurosci*. **37**, 1–9 (2014).
57. Beiglary, S. et al. Using wearable sensors to measure and predict personal circadian lighting exposure in nursing home residents: model development and validation. *JMIR Aging* **8**, e72338–e72338 (2025).
58. Patel, S., Park, H., Bonato, P., Chan, L. & Rodgers, M. A review of wearable sensors and systems with application in rehabilitation. *J. Neuroeng. Rehabil.* **9**, 21 (2012).
59. Danilenko, K. V. et al. Wearable light-and-motion dataloggers for sleep/wake research: a review. *Appl*. *Sci*. **12**, 11794 (2022).
60. Banasiak, M., Wilkerson, A. & Safranek, S. Evaluating occupant light exposure and usage patterns in an inpatient behavioral health unit. *Health Environ. Res. Des. J*. https://doi.org/10.1177/19375867221150226 (2023).
61. Stevenson, S., Suppiah, H., Ruddy, J., Murphy, S. & Driller, M. Higher levels of morning and daytime light exposure associated with positive sleep indices in professional team sport athletes. *Nat. Sci. Sleep.* **16**, 1279–1290 (2024).
62. VoPham, T., Ton, M. & Weaver, M. D. Spatiotemporal light exposure modeling for environmental circadian misalignment and solar jetlag. *Environ. Epidemiol.* **8**, e301 (2024).
63. Kelly, J. M. & Wilkerson, A. Comparing Occupant Light Exposure in Factory and Office Spaces. PNNL-34490, Pacific Northwest National Laboratory, Richland, WA https://doi.org/10.2172/2281871 (2023).
64. Figueiro, M., Hamner, R., Bierman, A. & Rea, M. Comparisons of three practical field devices used to measure personal light exposures and activity levels. *Light. Res. Technol.* **45**, 421–434 (2013).
65. Markvart, J., Hansen, Å. M. & Christoffersen, J. Comparison and correction of the light sensor output from 48 wearable light exposure devices by using a side-by-side field calibration method. *LEUKOS - J. Illum. Eng. Soc. N. Am*. **11**, 155–171 (2015).
66. Ostrin, L. A., Sajjadi, A. & Benoit, J. S. Objectively measured light exposure during school and summer in children. *Optom. Vis. Sci.* **95**, 332–342 (2018).
67. Price, L. L. A., Khazova, M. & O'Hagan, J. B. Performance assessment of commercial circadian personal exposure devices. *Light. Res. Technol*. **44**, 17–26 (2012).
68. Read, S. A., Collins, M. J. & Vincent, S. J. Light exposure and physical activity in myopic and emmetropic children. *Optom. Vis. Sci.* **91**, 330–341 (2014).
69. Shneor, E., Gordon-Shaag, A., Doron, R., Benoit, J. S. & Ostrin, L. A. Utility of the Actiwatch Spectrum Plus for detecting the outdoor environment and physical activity in children. *J. Optom.* **17**, 100483 (2024).
70. Price, L. L. A., Lyachev, A. & Khazova, M. Optical performance characterization of light-logging actigraphy dosimeters. *J. Opt. Soc. Am. A* **34**, 545 (2017).
71. Howell, C. M., McCullough, S. J., Doyle, L., Murphy, M. H. & Saunders, K. J. Reliability and validity of the Actiwatch and Clouclip for measuring illumination in real-world conditions. *Ophthalmic Physiol. Opt.* **41**, 1048–1059 (2021).
72. Hwang, Y., McPhillips, M. V., Talwar, S., Perez, G. A. & Hodgson, N. A. The relationship between light exposure before bedtime and daytime sleepiness among people living with cognitive impairment. *J. Appl. Gerontol*. **41**, 1686–1694 (2022).
73. Izci Balserak, B., Hermann, R., Hernandez, T. L., Buhimschi, C. & Park, C. Evening blue-light exposure, maternal glucose, and infant birthweight. *Ann. N. Y. Acad. Sci*. **1515**, 276–284 (2022).
74. Izmailova, E. S. et al. Evaluation of wearable digital devices in a phase I clinical trial. *Clin. Transl. Sci*. **12**, 247–256 (2019).
75. Kim, W. H. et al. The effect of bright light therapy on sleep and quality of life in patients with poststroke insomnia. *Psychosom. Med.* **84**, 123–130 (2022).
76. Mantua, J., Gravel, N. & Spencer, R. M. C. Reliability of sleep measures from four personal health monitoring devices compared to research-based actigraphy and polysomnography. *Sens. Switz.* **16**, 646 (2016).
77. Nagra, M., Rodriguez-Carmona, M., Blane, S. & Huntjens, B. Intra- and inter-model variability of light detection using a commercially available light sensor. *J. Med. Syst.* **45**, 46 (2021).
78. De Zambotti, M. et al. State of the science and recommendations for using wearable technology in sleep and circadian research. *SLEEP* **47**, zsad325 (2024).
79. Bhandari, K. R., Mirhajianmoghadam, H. & Ostrin, L. A. Wearable sensors for measurement of viewing behavior, light exposure, and sleep. *Sensors* **21**, 7096 (2021).
80. De Vries, S., Gkaintatzi-Masouti, M., Van Duijnhoven, J., Mardaljevic, J. & Aarts, M. Recommendations for light-dosimetry field studies based on a meta-analysis of personal light levels of office workers. *Light. Res. Technol.* **57**, 47–70 (2025).
81. Stefani, O. et al. Wearable light loggers in field conditions: corneal light characteristics, user compliance, and acceptance. *Clocks Sleep.* **6**, 619–634 (2024).
82. Ancoli-Israel, S. et al. The SBSM guide to actigraphy monitoring: clinical and research applications. *Behav. Sleep. Med.* **13**, S4–S38 (2015).
83. Hébert, M., Martin, S. K., Lee, C. & Eastman, C. I. The effects of prior light history on the suppression of melatonin by light in humans. *J. Pineal Res.* **33**, 198–203 (2002).
84. Smith, M. T. et al. Use of actigraphy for the evaluation of sleep disorders and circadian rhythm sleep-wake disorders: an american academy of sleep medicine clinical practice guideline. *J. Clin. Sleep. Med.* **14**, 1231–1237 (2018).
85. Swope, C. B. et al. Factors associated with variability in the melatonin suppression response to light: a narrative review. *Chronobiol. Int.* **40**, 542–556 (2023).
86. Zauner, J., Biller, A. M. & Spitschan, M. Impact of light logger and dosimeter placement on wearability and appeal in real-life settings. *Open Res. Eur.* **5**, 293 (2025).
87. Spitschan, M. et al. How to report light exposure in human chronobiology and sleep research experiments. *Clocks Sleep.* **1**, 280–289 (2019).
88. Spitschan, M. et al. ENLIGHT: a consensus checklist for reporting laboratory-based studies on the non-visual effects of light in humans. *eBioMedicine* **98**, 104889 (2023).
89. Guidolin, C. et al. Protocol for a prospective, multicentre, cross-sectional cohort study to assess personal light exposure. *BMC Public Health* **24**, 3285 (2024).
90. Bretschneider, E. Identification of the blue-yellow metameric locus of the circadian stimulus model. *Light. Res. Technol*. https://doi.org/10.1177/14771535251327396 (2025).
91. Truong, W., Trinh, V. & Khanh, T. Circadian stimulus – A computation model with photometric and colorimetric quantities. *Light. Res. Technol.* **52**, 751–762 (2020).
92. Schlangen, L., Belgers, S., Cuijpers, R., Zandi, B. & Heynderickx, I. Correspondence: designing and specifying light for melatonin suppression, non-visual responses and integrative lighting solutions – establishing a proper bright day, dim night metrology. *Light. Res. Technol.* **54**, 761–777 (2022).
93. Kore, R. & Durmus, D. Curve-fitting correction method for the nonlinear dimming response of tunable SSL devices. *Technologies* **11**, 162 (2023).
94. Kore R. & Durmus D. Performance of wearable light sensors for circadian photobiology. In *Proc. IES Annual Conference* (IES, 2023).

95. Xia, S., Mestas-Nuñez, A. M., Xie, H., Tang, J. & Vega, R. Characterizing variability of solar irradiance in San Antonio, Texas using satellite observations of cloudiness. *Remote Sens.* **10**, 2016 (2018).
96. Hönekopp, A. & Weigelt, S. Using light meters to investigate the light-myopia association – A literature review of devices and research methods. *Clin. Ophthalmol.* **17**, 2737–2760 (2023). **Volume**.
97. Willems, J. J. *Investigating the Performance and thea-opic Compatibility of the LightLog Photometer* (Eindhoven University of Technology, 2017).
98. Humphreys, M. A., Rijal, H. B. & Nicol, J. F. Updating the adaptive relation between climate and comfort indoors; new insights and an extended database. *Build. Environ.* **63**, 40–55 (2013).
99. Kingma, B. R., Frijns, A. J., Schellen, L. & Van Marken Lichtenbelt, W. D. Beyond the classic thermoneutral zone: Including thermal comfort. *Temperature* **1**, 142–149 (2014).
100. Mairiaux, P., Malchaire, J. & Candas, V. Prediction of mean skin temperature in warm environments. *Eur. J. Appl. Physiol.* **56**, 686–692 (1987).
101. Zhang, Q. et al. Characterization of temperature profiles in skin and transdermal delivery system when exposed to temperature gradients in vivo and in vitro. *Pharm. Res.* **34**, 1491–1504 (2017).
102. CIE. CIE Position statement on integrative lighting recommending proper light at the proper time. https://doi.org/10.25039/ps.b2twa77g (2024).
103. Spitschan, M. Opinion: Future-proofing circadian research. *Light. Res. Technol.* **51**, 818–819 (2019).
104. Durmus, D. Correlated color temperature: use and limitations. *Light. Res. Technol.* **54**, 363–375 (2022).
105. CIE. CIE 015:2018: Colorimetry, 4th edn. CIE Central Bureau, Vienna https://doi.org/10.25039/TR.015.2018 (2018).
106. ISO/CIE. ISO/CIE 19476:2014(E): Characterization of the Performance of Illuminance Meters and Luminance Meters. International Standard, 1st edn. ISO/CIE (2014).
107. Ying, S.-P., Chou, P. T. & Fu, H.-K. Influence of the temperature dependent spectral power distribution of light-emitting Diodes on the illuminance responsivity of a photometer. *Opt. Lasers Eng.* **51**, 1179–1184 (2013).

Acknowledgements

This work was supported by the Pennsylvania State University Institutes of Energy and Environment's (IEE) Seed Grant Program.

Author contributions

K.R. contributed to conceptualization, data curation, formal analysis, software, and writing original draft. B.V. contributed to the data curation and formal analysis. R.D. contributed to the conceptualization, helped with resources, reviewing, and editing the drafts. M.J. contributed to the conceptualization, helped with resources, reviewing, and editing the drafts. C.A.M. also contributed to the conceptualization, helped with funding acquisition, project administration, supervision, reviewing, and editing the drafts. D.D. contributed to conceptualization, funding acquisition, project administration, supervision, reviewing, and editing the drafts.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to R. S. Kore.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026